

TESTS FOR INDEPENDENCE (CHI SQUARED TESTS.)

The Chi squared test always gives a positive result because it involves squared observations. As In t-distribution, there are many chi-squared distributions each with a different shape that depends on the number of degrees of freedom. The number of degrees of freedom is determined by the number of categories minus one, that is $k-1$ and NOT by the size of the sample. If a sample of 150 undergraduates in a university are categorised as First year, second year, third year and fourth year, there would be

$K-1=4-1=3$ degrees of freedom.

The following are properties of the chi-squared distribution:

1. Chi-square is non negative i.e. it is either 0 or positive
2. The chi-square distribution is not symmetrical. Its skewness is positive. However, as the number of degrees of freedom increases, chi-square approaches a symmetric distribution.
3. There is a family of chi square distributions. There is a particular distribution for each degree of freedom.

Goodness-of-fit test

The chi-squared distribution is used to find out how well an observed set of frequencies, f_o fit an expected set of frequencies, f_e

Steps in test for χ^2 -goodness of fit:

The five step hypothesis testing procedure followed earlier applies:

Step 1: The null and alternative hypotheses are stated. The null hypothesis, H_0 : there is *independence* and the alternative hypothesis H_a : there is *no independence*

Step 2. Select level of significance. If not stated take the level of significance =0.05. Recall that this is the probability of making Type I error. It means that the probability is 0.05 of incorrectly rejecting a true hypothesis.

Step 3. Use the chi squared test statistic given by

$$\chi^2 = \sum \frac{(f_o - f_e)^2}{f_e}$$

where f_0 is an observed frequency in a category

f_e is an expected frequency in a particular category. For a contingency table we use

$$f_e = \frac{\text{row_total} * \text{column_total}}{\text{grand_total}}$$

Step 4. Obtain the critical value of χ^2 with appropriate degrees of freedom (df)

Step 5. Make a decision: Reject the null hypothesis whenever the calculated value of χ^2 is greater than the tabulated value of χ^2

Example 1.

A sample of people were asked by a research institution to rate the honesty and ethical standards of various groups. The results are shown below:

Professions	High	Average	Low
Medical Doctors	23	35	10
Lawyers	24	43	27
Business Executives	18	55	20
Politicians	14	43	38

Use the 0.01 level of significance to test whether all professions are rated similarly.

Example 2.

Five different brands of tea were examined for quality. A total of 24 cans of each brand were checked. The numbers of acceptable and unacceptable cans are as follows.

Quality	Brands				
	A	B	C	D	E
Acceptable	3	10	5	3	9
Unacceptable	21	14	19	21	15

At 0.05 significance level, can we conclude the five brands are of the same quality?

Example 3.

A manufacturer of women garments is interested in determining if age is a factor in whether women believe they would buy a particular garment. Accordingly, the firm surveyed three age groups and asked each person to rate the garment as either excellent, average or poor. The results are shown below.

Rating	Age Group			
	<i>15 to 24</i>	<i>25 to 39</i>	<i>40 to 55</i>	<i>Total</i>
Excellent	40	47	46	133
Average	51	74	57	182
Poor	29	19	37	85